

SAIKUNDAN SUDDAPALLI

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Summary

Master's in Computer Science with expertise in Machine Learning, Cybersecurity, and C++ development. Proven ability to build intelligent systems, evidenced by projects like the Kundan Job Assistant and research in Quantum Neural Networks. Skilled in deep learning (TensorFlow, PyTorch, LLMs), reverse engineering, and low-latency system development

Education

University of North Georgia

M.S. in Computer Science

GPA: 4.0/4.0

Jan 2024 – Dec 2024

Dahlonega, GA

Thesis: Quantum Neural Networks on Imbalanced Datasets

Jawaharlal Nehru Technological University

B.E. in Computer Science

GPA: 3.0/4.0

July 2017 – Aug 2021

Hyderabad, India

Experience

Graduate Research Assistant

University of North Georgia

May 2024 – Dec 2024

Dahlonega, GA, USA

- Contributed to the development of an advanced version of the CrystaLLM library for large language models, focusing on interpretability and explainability, to aid in the discovery of novel therapeutic compounds for Alzheimer's disease. Led research in training deep models (e.g., GPT-based architecture) to predict chemical properties, accelerating drug discovery processes.
- Collaborated with a multidisciplinary research team from Texas A&M University, University of Georgia (UGA), and other institutions, representing the Computer Science Department.
- Developed and optimized CrystaLLM using PyTorch, focusing on self-supervised learning techniques to enhance model performance.
- Fine-tuned large language models (LLMs) with LoRA (Low-Rank Adaptation) to improve efficiency and effectiveness in drug discovery tasks.
- Contributed to the integration of chemical and physical data from various departments to enhance the model's predictive capabilities.
- Conducted experiments and analyzed results to validate the model's performance and make data-driven decisions. Documented research findings and presented progress updates to the research team and stakeholders.
- Utilized LoRA to fine-tune existing open-source LLM models (e.g., LLaMA, GEEMA) for specific chemical tasks.
- Utilized high-performance computing resources (HPC) on Bridges to expedite training and inference.
- Developed and led tutorials on Quantum Computing using IBM Quantum Experience and Qiskit.

C++ & Python Developer

Tata Consultancy Services

Nov 2021 – Nov 2023

Hyderabad, Telangana, India

- Developed and customized a trading platform for a client in the BFSI sector, specifically the Multi Commodity Exchange (MCX), utilizing extensive experience in C/C++ and Python, with a strong focus on UNIX platforms and proficiency in analyzing and debugging in C, C++, and Python.
- Demonstrated expertise in C++ development, enhancing the trading platform by mastering critical components such as the Order Book Interface, Market Data Distribution, Info Feed, and integrating seamlessly with Kafka, resulting in enhanced analytical prowess through meticulous C++ craftsmanship.
- Achieved a significant 1 million user increase by optimizing the Enhanced Order Book Interface, significantly growing user engagement and market footprint, showcasing ability to drive business growth through technical expertise.
- Engineered a low-latency Info Feed for alpha clients, surpassing expectations and elevating satisfaction levels, highlighting ability to deliver high-quality solutions under tight deadlines.
- Implemented a seamless Git flow, enhancing collaboration efficiency by 30% and achieving error-free code deployments with improved version control, demonstrating expertise in DevOps and version control systems.
- Developed a robust CI/CD pipeline using Bash, Python, and Ansible, reducing deployment time by 40% and improving deployment reliability by 25%, showcasing ability to automate and streamline development processes.
- Led a Python-powered deployment process that cut deployment time by 50% across 100 servers, enhancing efficiency and time utilization, highlighting ability to lead and manage large-scale deployments.

- Ensured a flawless environment handover well before live trading, resulting in a seamless transition with zero disruptions, demonstrating ability to manage and deliver high-stakes projects.
- Automated Python scripts for monthly SLA report extraction, reducing report generation time by 30%, showcasing ability to automate and optimize business processes.
- Automated system checks, reducing manual effort by 20% and fostering a proactive and reliable system, highlighting ability to identify and implement process improvements.

Full Stack Developer

Apr 2020 – Oct 2021

River Bend Data Solutions

Hyderabad, Telangana, India

- Specialized in developing cutting-edge ML-based analytics for wearable devices.
- Contributed to the development of a medical chatbot assistant by creating algorithms to process data from band watches, focusing on heart rate time series analysis.
- Improved model accuracy by 10% through rigorous data preprocessing and feature engineering.
- Integrated ML models into complex systems and contributed to frontend development using ReactJS to visualize key metrics.

Certifications

- TensorFlow Certified Developer, Google
- Azure Administrator Associate (Exam 104), Microsoft

Technical Skills

Languages: C++ Python Java JavaScript C SQL Assembly Bash

Machine Learning: TensorFlow PyTorch LLMs Scikit-learn NLP LORA

Cybersecurity: Malware Analysis Reverse Engineering

Tools: Ghidra Docker Git Kafka GDB cmake Unity

Projects

ATS killer: Your AI-Powered Resume Tailoring Assistant

GitHub

C++ Python llama.cpp Ollama LLMs threading Automation Docker

- ATS killer is a dynamic application designed to streamline and enhance the resume tailoring process. Initially conceived with a cloud-based LLM API, which proved time-intensive, ResumeForge evolved into a powerful, locally-executed solution
- **Chrome Extension Frontend:** Provides a seamless user interface for job description input.
- **Local LLM Processing:** Leverages a highly efficient local language model (Llama, with ongoing development using optimized Mistral models) implemented in C++ and Python, significantly reducing processing time from 25-30 minutes to 2-3 minutes per resume.
- **Flask Middleware:** Ensures smooth data transfer between the extension and the LLM.
- **Data Persistence:** Utilizes CSV files for data logging and future analysis, and excel files for content insertion.
- **Intelligent Resume Enhancement:** Analyzes job descriptions to strategically augment resume content, improving match relevance without altering the user's core experience.
- **Threaded C++ Optimization (Under Development):** Current efforts are focused on implementing a multi-threaded C++ backend to further accelerate resume section-by-section optimization.
- Transitioned from a slow cloud-based LLM API (Mistral) to a fast local LLM (Llama, and optimized Mistral) to improve speed and reduce API limitations. Achieved a substantial reduction in resume processing time. Developed a robust system for local LLM integration and data management.
- Currently working on increasing the speed of the local LLM using multithreading with C++.

Technical Portfolio

- Banking trogen Malware analysis
- Quantum-on-imbalanced-data
- FakeVideoDetector
- OS-threading
- Ubuntu Multipass
- LLM vs DNNs